

Conjugate-Paired Neural Demodulation for OFDM With Receiver I/Q Imbalance

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Abstract—In-phase/quadrature (I/Q) imbalance (IQI) distorts orthogonal frequency-division multiplexing (OFDM) signals by mixing each subcarrier with its mirror. To improve neural demodulation under IQI, this study proposes a processing structure that estimates IQI from the received packet, compensates the mixing before channel equalization, and supplies mirror-paired samples to the network. The IQI estimator and bit detector are trained jointly using transmitted bit labels. Comparisons use linear detectors and direct neural models without IQI compensation or explicit mirror pairing. For quadrature phase-shift keying (QPSK), the structure reduces mean nonideal-IQI bit error rate (BER) by 8.85–10.05% relative to direct models across convolutional neural network (CNN), Transformer, and U-Net backbones. Extending the structure to 16-ary quadrature amplitude modulation (16-QAM) with a CNN reduces BER by 14.34%. These simulations supply the channel response and noise variance to each receiver.

Code availability: The implementation is available at the project repository.

Index Terms—I/Q imbalance, neural demodulation, OFDM

I. INTRODUCTION

IN-PHASE (I) and quadrature (Q) branches represent the two components of a received complex signal. Differences in their analog gains or deviations from their intended 90° phase separation cause I/Q imbalance (IQI) [1], [2]. IQI adds a conjugated copy of the signal to the desired signal; conjugation reverses the sign of its imaginary component.

Orthogonal frequency-division multiplexing (OFDM) carries symbols on separate frequency subcarriers [3]. Receiver IQI mixes each subcarrier with the conjugate of the subcarrier at the opposite frequency, called its mirror [4], [5]. This interference can cause bit errors even when background noise is weak [5].

Traditional receivers estimate IQI parameters and use them to remove the unwanted mixing [1], [5]. Some estimate IQI from received-signal statistics without known training symbols [6]. These methods motivate including an IQI model in neural receivers. Neural receivers learn detection from received samples [7]–[9], including OFDM signals with nonlinear distortion [10] and real-time implementations [11]. Hybrid designs combine learned and conventional processing [12], [13], with online learning to adapt receivers to changing channels [14]. A network can learn relationships between subcarriers, but a general input layout does not explicitly pair mirror samples. Conventional channel equalization corrects channel amplitude and phase changes while leaving IQI mixing in place. This study therefore combines learned IQI estimation and compensation with explicit mirror-paired inputs to improve neural demodulation under IQI.

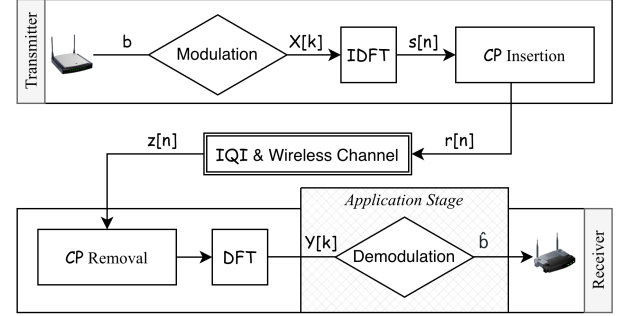


Fig. 1. OFDM link. The combined channel/IQI block contains, in order, the wireless channel, additive noise, and receiver IQI. The demodulation stage includes IQI compensation (proposed only), channel equalization, and bit detection. The symbol index t is omitted in the diagram.

This study tests the proposed structure with three backbones (main networks for bit detection): a convolutional neural network (CNN), a Transformer, and a U-Net. Comparisons use IQI-unaware and true-IQI linear references and a direct neural model for each backbone. Direct models retain channel equalization and matched backbone and training settings, but omit IQI estimation, compensation, and mirror pairing. With known channels and noise variances, the structure reduces mean nonideal-IQI BER by 8.85–10.05% for QPSK across the three backbones and by 14.34% for CNN 16-QAM.

II. SIGNAL MODEL AND RECEIVER DESIGN

A small multilayer perceptron (MLP) estimates IQI. Compensation, channel equalization, mirror pairing, and neural bit prediction follow. Figure 2 shows its front end.

A. From Transmitted Bits to Received Samples

Figure 1 shows the simulated link. Random bits are mapped to QPSK or 16-QAM symbols $X_t[k]$, where t indexes OFDM symbols and k indexes subcarriers. An inverse discrete Fourier transform (IDFT) converts them to time samples. A cyclic prefix (CP), which copies the end of each symbol to its beginning, is then inserted [15]. The signal passes through the wireless channel, additive noise, and receiver IQI, in that order. The receiver removes the CP and applies a DFT to obtain $Y_t[k]$ [5], [15].

To describe IQI, let $r_t[n]$ denote a noisy time sample just before the imbalanced receiver branches. Their output is [4], [5]

$$z_t[n] = \mu r_t[n] + \nu r_t^*[n]. \quad (1)$$

Here, $(\cdot)^*$ denotes complex conjugation. The coefficient μ scales the original signal, while ν sets the unwanted conjugate contribution. The coefficients are calculated from gain mismatch g in decibels and phase mismatch ϕ in radians:

$$\mu = \frac{1 + ae^{-j\phi}}{\sqrt{2(1+a^2)}}, \quad \nu = \frac{1 - ae^{j\phi}}{\sqrt{2(1+a^2)}}, \quad a = 10^{g/20}. \quad (2)$$

The quantity a is the I/Q amplitude ratio. For the independently drawn, equiprobable QPSK/16-QAM symbols and circular Gaussian noise used here, $\mathbb{E}[r_t[n]^2] = 0$, so the cross terms in output power vanish [16]. Hence, $\mathbb{E}|z_t[n]|^2 = (|\mu|^2 + |\nu|^2)\mathbb{E}|r_t[n]|^2$. The normalization sets $|\mu|^2 + |\nu|^2 = 1$, preserving average power while IQI changes the signal shape and mirror contribution. Ideal branches have $g = \phi = 0$, giving $\mu = 1$ and $\nu = 0$. The simulation holds g and ϕ fixed within each packet to model a receiver mismatch that changes slowly relative to the packet duration.

For the frequency-domain description, define the observation that would be obtained without IQI as [15]

$$R_t[k] = H[k]X_t[k] + W_t[k], \quad H[k] = \sum_{\ell=0}^{L-1} h[\ell]e^{-j2\pi k\ell/N}. \quad (3)$$

Here, N is the DFT size and $h[\ell]$ weights the channel path delayed by ℓ samples. A CP of at least $L - 1$ samples prevents interference from the preceding symbol; the DFT then reduces the channel to one complex multiplier $H[k]$ per subcarrier [15]. The simulation keeps the channel fixed within each packet. Additive white Gaussian noise (AWGN) gives independent samples $W_t[k]$, whose real and imaginary parts each have zero mean and variance $\sigma_w^2/2$. Since $|W|^2 = (\text{Re } W)^2 + (\text{Im } W)^2$, $\mathbb{E}|W_t[k]|^2 = \sigma_w^2$, the pre-IQI noise power per subcarrier. Supplying $H[k]$ and σ_w^2 to every receiver allows a controlled comparison without channel- or noise-estimation errors.

Conjugating time samples reverses their frequency indices. Consequently, the actual DFT output is [5]

$$Y_t[k] = \mu R_t[k] + \nu R_t^*[\bar{k}], \quad \bar{k} = (-k \bmod N). \quad (4)$$

For example, bins 1 and 63 form a mirror pair when $N = 64$. The second term brings both signal and noise from the mirror into bin k . $R_t[k]$ is the pre-IQI reference, not an extra measured input.

B. Estimating and Removing IQI

The estimator in Fig. 2 uses four values computed from the packet and the supplied noise level. It first measures the average received power P and the normalized product ρ of mirror samples:

$$P = \frac{1}{T|\mathcal{K}|} \sum_{t=0}^{T-1} \sum_{k \in \mathcal{K}} |Y_t[k]|^2, \quad \rho = \frac{1}{T|\mathcal{K}|P} \sum_{t=0}^{T-1} \sum_{k \in \mathcal{K}} Y_t[k]Y_t[\bar{k}]. \quad (5)$$

Here, T is the number of OFDM symbols per packet and $\mathcal{K}$ is the set of active subcarriers. The simulation uses $T = 14$ and $\mathcal{K} = \{1, \dots, 26, 38, \dots, 63\}$, containing 52 subcarriers and all their mirrors.

Without IQI, independent zero-mean symbols and noise give zero expected product between distinct mirror observations. IQI makes each observation contain part of its mirror, so their product can have a nonzero average. This product is taken without conjugation, as used in complex-signal statistics [16]. Dividing by P removes the overall power scale. The five real inputs are

$$\mathbf{s}_{\text{IQI}} = [\text{Re } \rho, \text{Im } \rho, |\rho|, \log_{10} P, \log_{10} \sigma_w^2]^T. \quad (6)$$

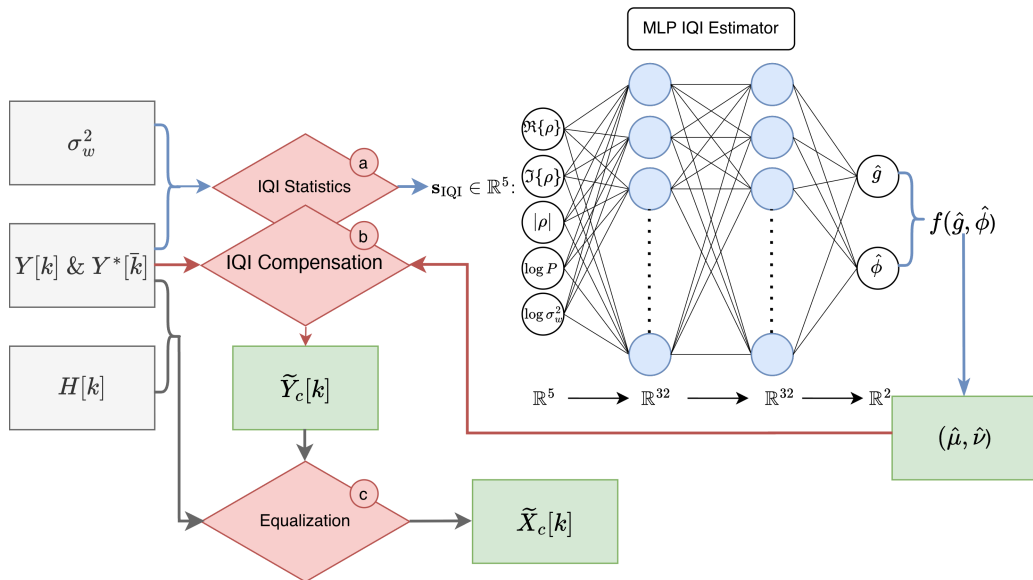


Fig. 2. Proposed front end. Five numbers summarize the received packet and its noise level. An MLP estimates gain and phase mismatch; (2) converts these estimates to mixing coefficients. Compensation removes the estimated mirror mixing before channel equalization. The equalizer uses $\tilde{Y}_c$, H , and σ_w^2 ; mirror pairing and neural bit detection follow.

The inputs describe mirror dependence, received power, and noise level.

The MLP, a small fully connected network, maps these five inputs to estimated gain and phase mismatch ($\hat{g}, \hat{\phi}$). It has two hidden layers of 32 units with rectified linear unit (ReLU) activations. Scaled hyperbolic-tangent outputs limit the estimates to ± 6 dB and $\pm 30^\circ$. The phase is converted to radians, and (2) gives $(\hat{\mu}, \hat{\nu})$ for the whole packet.

Using a signal and its conjugate together is established in complex-signal estimation [17]. Here, compensation combines the two mixed observations to cancel the mirror term [1], [5]. Substituting (4) for both members of the pair gives

$$\mu^* Y_t[k] - \nu Y_t^*[k] = (|\mu|^2 - |\nu|^2) R_t[k].$$

Thus, subtraction cancels the unwanted component, and division by the remaining scale recovers the pre-IQI signal. With estimated coefficients, the compensated signal is

$$\tilde{Y}_{c,t}[k] = \frac{\hat{\mu}^* Y_t[k] - \hat{\nu} Y_t^*[k]}{\max\{|\hat{\mu}|^2 - |\hat{\nu}|^2, 10^{-3}\}}. \quad (7)$$

The lower bound prevents division by a value near zero. Channel equalization then corrects channel amplitude and phase while limiting noise amplification on weak subcarriers:

$$\tilde{X}_{c,t}[k] = \frac{H^*[k] \tilde{Y}_{c,t}[k]}{|H[k]|^2 + \sigma_w^2}. \quad (8)$$

This is the scalar linear minimum mean-square error (LMMSE) equalizer for an IQI-free channel [9]. Figure 3 illustrates the two corrections together: after IQI compensation and channel equalization, the samples lie closer to the transmitted constellation points.

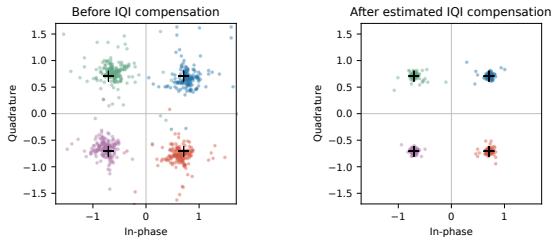


Fig. 3. One QPSK test packet at $E_s/N_0 = 30$ dB and $(g, \phi) = (2 \text{ dB}, 10^\circ)$. Left: channel equalization alone. Right: IQI compensation followed by the same equalizer. Symbol mean-squared error decreases from 0.1059 to 0.00507. Crosses mark ideal symbols; colors identify transmitted bit pairs.

C. Neural Bit Detection and the Direct Model

After compensation and equalization, the proposed structure places each sample beside its mirror conjugate. This lets the detector use both samples to handle remaining mirror interference. Its input at each location is

$$\mathbf{u}_t[k] = [\text{Re}\{\tilde{X}_{c,t}[k]\}, \text{Im}\{\tilde{X}_{c,t}[k]\}, \text{Re}\{\tilde{X}_{c,t}^*[k]\}, \text{Im}\{\tilde{X}_{c,t}^*[k]\}, \log_{10} \sigma_w^2]^T. \quad (9)$$

The current sample, mirror conjugate, and noise level form a $5 \times 14 \times 52$ input grid.

The direct model omits the IQI estimator, compensation, and explicit mirror pairing. The same channel equalizer acts directly on $Y_t[k]$:

$$\tilde{X}_t[k] = \frac{H^*[k] Y_t[k]}{|H[k]|^2 + \sigma_w^2}. \quad (10)$$

Its inputs are $\text{Re} \tilde{X}_t[k]$, $\text{Im} \tilde{X}_t[k]$, and $\log_{10} \sigma_w^2$, giving a $3 \times 14 \times 52$ grid. The network sees the full grid, but mirror samples are not explicitly paired.

Both models use a 3×3 input convolution with ReLU to produce 64 features per location. The backbone processes this grid and returns the same grid size. The evaluation uses residual CNN blocks [18], Transformer encoders [19], and a U-Net [20]; Table I gives their settings and Fig. 4 shows the CNN. The CNN and U-Net use group normalization [21]; the Transformer uses layer normalization [22] to rescale internal features. A 1×1 output convolution produces $Q = \log_2 M$ bit scores per M -ary symbol. A positive score predicts bit 1; otherwise, it predicts bit 0.

For training, each score ℓ_i is converted to a probability $p_i = 1/(1 + e^{-\ell_i})$. Both models minimize binary cross-entropy [9]:

$$\mathcal{L} = -\frac{1}{N_b} \sum_{i=1}^{N_b} [b_i \log p_i + (1 - b_i) \log(1 - p_i)], \quad (11)$$

where b_i is the transmitted bit and N_b counts bits in the training batch. The loss penalizes incorrect bit probabilities. For the proposed structure, gradients pass through equalization and compensation to train the MLP together with the bit detector. No separate symbol, gain, or phase targets are used. The two models are trained separately with matched backbone settings and training procedures.

III. EXPERIMENTS

A. Dataset and Comparisons

The simulation generates labeled packets using the time-domain link in Fig. 1. Independent random bits are mapped to unit-average-energy QPSK or Gray-labeled 16-QAM, followed by IDFT and CP insertion. Each packet draws a new three-tap channel, noise realization, and IQI parameters; the channel and IQI remain fixed across its 14 symbols. Each channel tap has independent zero-mean Gaussian real and

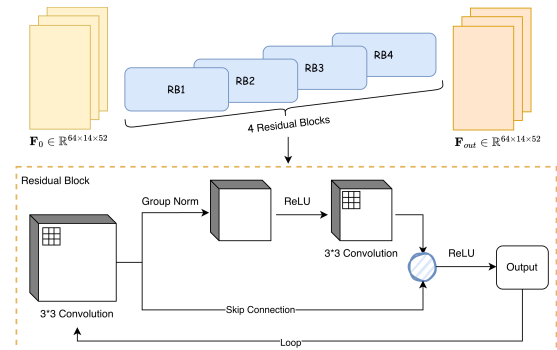


Fig. 4. CNN backbone. Four residual blocks (RBs) preserve the $64 \times 14 \times 52$ feature-grid size.

TABLE I
BACKBONES AND TRAINABLE PARAMETER COUNTS FOR QPSK
DEMULATORS

Backbone	Configuration	Trainable parameters	
		Proposed	Direct
CNN	Four 64-channel residual blocks (Fig. 4).	300,836	298,370
Transformer	Two 64-wide, four-head encoder layers with learned positions and 128-unit feed-forward layers.	118,052	115,586
U-Net	32-channel encoder, half-resolution path, two residual blocks, and additive skip connection.	116,036	113,570

imaginary parts and average power 1/3. Noise is added before IQI. The receiver then removes the CP and applies the DFT. Table II lists the settings.

TABLE II
SIMULATION AND TRAINING SETUP

Item	Setting
OFDM grid	$N = 64$, bins 1–26 and 38–63; 14 symbols
Cyclic prefix / channel	16 samples / 3 equal-power Rayleigh taps
Symbol-energy/noise ratio E_s/N_0	0.5:30 dB; AWGN before receiver IQI
Training IQI	$g \sim \mathcal{U}[-2, 2]$ dB, $\phi \sim \mathcal{U}[-10, 10]^\circ$
Ideal fraction	20% of training packets
Data split	409,600 packets/modulation; 80%/20%
Optimization	AdamW [23], 10^{-3} , batch 64, 10 epochs
Test size	512 packets per condition and SNR value

With unit symbol energy, pre-IQI $E_s/N_0 = 1/\sigma_w^2$ per active subcarrier, excluding CP and inactive-subcarrier overhead. Direct/proposed pairs share the packet-disjoint training/validation split and initialization seed. One model is trained per configuration; the checkpoint with the lowest validation BER is tested.

Independent test packets share bits, channels, and pre-IQI noise across receivers and IQI conditions. The eight (g [dB], ϕ [degrees]) settings are (0, 0), (± 2 , 0), (0, ± 10), (2, 10), (-2 , -10), and (3, 15); the last lies outside the training range.

The IQI-unaware linear detector selects the nearest constellation point to $Y_t[k]/H[k]$, equivalent to (10) after correcting its amplitude shrinkage. The true-IQI reference first removes IQI with the simulated μ, ν , then applies the same detector. Neural receivers receive no true IQI parameters.

B. Results

Table III averages BER equally over the seven nonideal IQI conditions and seven SNR values. Relative reduction is $(\text{BER}_{\text{direct}} - \text{BER}_{\text{proposed}})/\text{BER}_{\text{direct}}$. The structure lowers mean BER by 8.85–10.05% across QPSK backbones and by 14.34% for CNN 16-QAM. Each configuration improves all 49 nonideal condition–SNR points.

TABLE III
CROSS-BACKBONE NONIDEAL-IQI BER

Setting	Direct	Proposed	Reduction
CNN / QPSK	0.06418	0.05773	10.05%
Transformer / QPSK	0.06338	0.05777	8.85%
U-Net / QPSK	0.06417	0.05782	9.90%
CNN / 16-QAM	0.12457	0.10671	14.34%

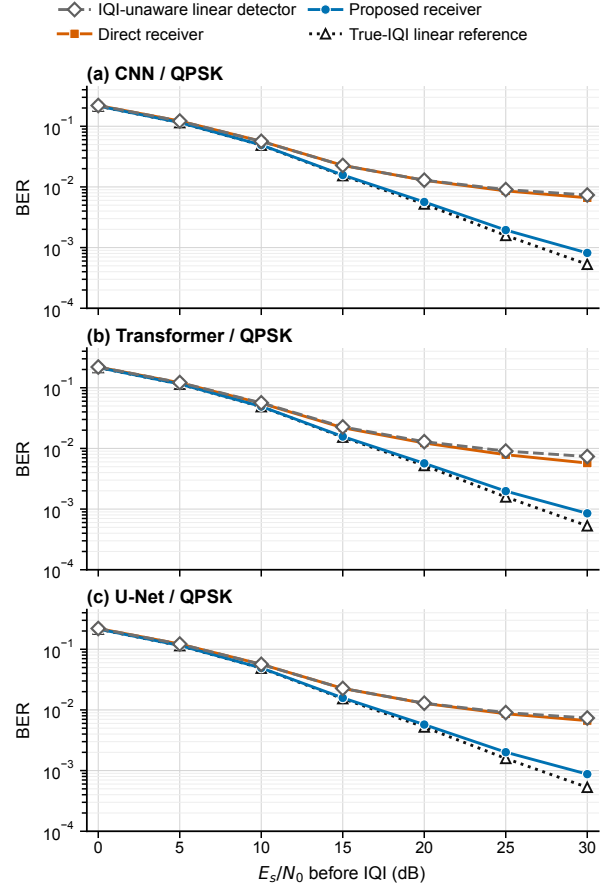


Fig. 5. QPSK BER versus pre-IQI E_s/N_0 , averaged equally over seven nonideal IQI conditions. Each condition–SNR pair uses 512 matched test packets.

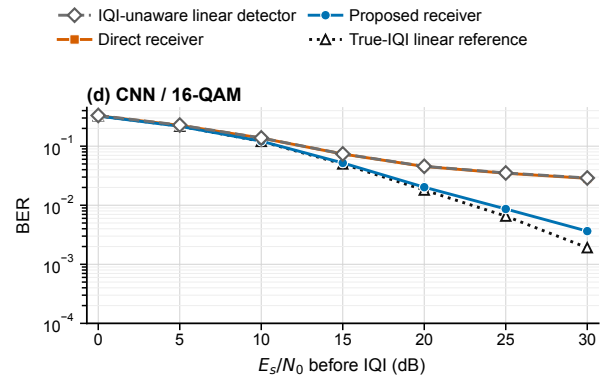


Fig. 6. CNN 16-QAM BER under the same matched test protocol as Fig. 5.

Figures 5 and 6 show similar BER at low SNR. From 15 to 30 dB, the direct and IQI-unaware linear curves flatten, while the proposed curves decrease toward the true-IQI reference. The three QPSK backbones show similar gains; CNN 16-QAM follows the same trend at higher BER. A gap to the reference remains at high SNR. Future work will evaluate how channel and noise estimation errors affect compensation and detection, and test the structure on hardware with frequency-dependent or time-varying IQI. These tests will assess whether the simulated gains persist under practical receiver conditions.

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